

BIO401



VU Expert B.S Zoologists
(All Semesters)

Biostatistics

MID TERM

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By

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VU EXPERT ZOOLOGISTS

Subjective Questions

Question no 01:

Define biostatistics:

The use of statistical methods to solve the biological problems. It is also called biometry.

Question no 2:

Population and Sample:

Population refers to the entire group of individuals that share a common characteristic and are the focus of a study at a particular time. In population the measurable quantity is called Parameter.

While Sample is a subset of the population that is selected for the actual analysis. Sampling is Often done when it is impossible to study the entire population. In sample the measurable quantity is called statistic.

Question no.: 3

Hypothetical Population?

A hypothetical population is a population that is imaginary or doesn't exist in a solid form. It is the opposite of an existing population. Here are some examples of hypothetical populations:

1. The outcomes of an experiment that is repeated infinitely
2. The outcomes of tossing a coin infinitely, resulting in a hypothetical population of "heads" and "tails"

Question no:4

Note on Survey?

Surveys

A method to collect direct feedback and information from a target audience. Surveys can be conducted online, over the phone, or face-to-face.

Question no 5:

Finite Population (Countable):

- The finite population is also known as a countable population in which the population can be counted.
- For statistical analysis, the finite population is more advantageous.
- Example: employees of a company.

Infinite Population (Uncountable):

- The population in which the counting of units in the population is not possible.
- It is less advantageous in statistical analysis
- Example: Number of germs in the patient's body.

Question no 6:

Difference between Parameter and Statistic?

A parameter is a numerical characteristic of a population. It is a fixed value that describes some aspect of the population. For example, the average blood pressure of all adults in a specific region is a parameter.

A statistic is a numerical characteristic of a sample. It is used to estimate or infer the Corresponding parameter of the population. For instance, the average blood pressure of a sample of 100 adults from the same region is a statistic.

Question no 7:

Detail note on Data?

Data refers to information collected, observed, or measured from population or sample during the study. This information can take various forms, including numerical values, categorical labels, or descriptive details, and it serves as the foundation for statistical analysis.

Biostatisticians use data to draw conclusions and derive meaningful insights about biological phenomena, health outcomes, or medical interventions.

There are two types of data:

1. Primary Data:

- This refers to data that is collected firsthand directly from the source.

- Common methods of collecting primary data include surveys, interviews, experiments, and Observations.
- Primary data is original, providing a high degree of relevance and specificity to the research question.
- It is expensive and take more time.

2. Secondary Data:

- The that has already been collected by someone else for a different purpose.
- Researchers utilize existing sources, such as published studies, government reports, or datasets, to obtain secondary data
- Secondary data is less specific to the researcher's needs and may not perfectly align with the study objectives.
- It is not expensive and time-saving way to access a large volume of information.

Question no 8:

Detail note on Variable?

A variable is a characteristic that can be measured and that can assume different values.

There are two types of variable:

1. Qualitative variable:

Qualitative variables are non-numeric and describe qualities or characteristics. Examples include Gender, blood type, place of birth, ethnic group, type of drug etc.

2. Quantitative variable:

A quantitative variable is one that can be measured and expressed numerically.

They have further two types:

- a) **Discrete Variable:** Discrete variables are numeric variables that can only take specific, distinct

Values, often counted in whole numbers and cannot be subdivided into smaller units.

Examples include the number of patients in a clinic, the number of red blood cells.

- b) **Continuous Variable:** Continuous variables are numeric variables that can take any value within

A range and can be measured with a high level of precision. These variables can be further Subdivided into smaller and smaller units, and they are often associated with measurements Rather than counts. Examples include height, weight, blood pressure, or temperature.

Question no 9:

Write detail note on Sampling?

The process of selecting a subset from a larger population for study. There are two types:

1. Probability sampling:

In this type random selection take place and every element of the population has equal chance to Be the part of selected sample. There many ways to do this:

a) Simple Random Sampling:

Every individual in the population has an equal chance of being selected, and each selection is Independent of others.

b) Stratified Random Sampling:

The population is divided into subgroups or strata based on certain characteristics, and then Random samples are taken from each stratum.

c) Cluster Sampling:

The population is divided into clusters, and then a random sample of clusters is selected. All Members within the chosen clusters are included in the sample.

- Stratified sampling, which involves selecting from every subgroup, cluster sampling involves Selecting entire groups.

d) Systematic Sampling:

- A random starting point is selected, and then every kth element in the population is chosen for the sample.

e) Multistage Sampling:

Involves multiple stages of sampling. It often combines various sampling methods; such as cluster sampling followed by simple random sampling within selected clusters.

2. Non- Probability Sampling:

The sampling technique where not every member of the population has an equal chance of being Included in the sample.

Some common types of non-probability sampling methods include:

a) Convenience Sampling:

Participants are selected based on availability.

b) Purposive Sampling:

Choose based on the purpose of the study.

c) Quota Sampling:

Choose on the base of some pre-set standard. Proportion of the sample should be same as Population.

d) Snowball Sampling:

First participants give the future participants. An initial set of participants is selected through purposeful sampling. These participants are then asked to refer others who might meet the study criteria.

Question no 10:

Purpose of Random sampling?

The purpose of random sampling is to select a subset of a population where every member has an equal chance of being chosen, ensuring that the sample accurately represents the larger population and minimizing the risk of bias in research results by eliminating the researcher's ability to influence the selection.

Random sampling is a probability sampling method that involves selecting a subset of individuals from a larger population in a way that gives every member an equal chance of being chosen.

Question no 11:

Types of Probability sampling?

The four main types of probability sampling are:

Simple random sampling

Each member of a population has an equal chance of being selected for the sample. This method ensures that the sample is representative of the population.

Systematic sampling

Researchers select members of the population at regular intervals, such as every 15th person on a list. This method can be useful when records of the target population already exist.

Stratified sampling

Researchers divide the population into subgroups based on shared characteristics, such as race, gender, or educational attainment. Then, each subgroup is randomly sampled using another probability sampling method.

Cluster sampling

Researchers divide the population into clusters, such as neighborhoods, cities, or regions. Then, they randomly select some of these clusters to serve as the sample. This method is useful for studying large populations that are spread out over a wide area. **Question no 12 :**

Why need sampling? Details on sample? Why there is need to get sample instead of whole population?

Answer:

In statistics, sampling is necessary because it allows researchers to collect data from a manageable subset of a population, which is often too large to study entirely, thus enabling them to make inferences about the whole population while saving time, cost, and effort by only analyzing a smaller representative sample.

Question no 13:

Classification of Sampling?

Classification or types of sampling are followed:

1. Simple random sampling
2. Systematic sampling
3. Stratified sampling
4. Cluster sampling
5. Convenient sampling
6. Purposive sampling
7. Quota sampling
8. Judgement sampling

Depends on marks you can further give details on these types.

Question no 14:

Importance of sampling trait?

Sampling is an important statistical technique because it allows researchers to draw conclusions about a population by studying a representative subset of it:

Efficiency

Sampling is faster and less expensive than collecting data from the entire population.

Practicality

Sampling is necessary when it's not possible to access or analyze the entire population, such as with historical records or large databases.

Destructive testing

Sampling is essential in destructive testing, such as in product testing or medical trials, to avoid excessive waste.

Quality of results

The more representative the sample is of the population; the more confident researchers can be in the quality of their results.

Market research

Sampling can help researchers understand what customers expect and what features they'd prefer in a new product.

Question no 15:

Give relevant statement that Convenient sampling is accurate or not?

Don't use convenience sampling when you need to be certain your sample is an accurate representation of your target audience. Convenience sampling focuses on getting research participants quickly and easily, so you can't be sure your sample is representative of your whole audience. While on the other hand, Probability sampling is a more accurate method in determining the true characteristics of the population.

Question no 16:

Central tendency?

Central tendency is defined as “the statistical measure that identifies a single value as representative of an entire distribution.” There are three main measures of central tendency: mode. Median. Mean.

In statistics, a central tendency is a central or typical value for a probability distribution.

Question no 17:

What is relationship of Mean, Median and Mode in term of Mean positively skewed distribution?

Answer:

In a positively skewed distribution, the mean is always greater than the median, which is always greater than the mode: $\text{mean} > \text{median} > \text{mode}$.

Positively Skewed Distribution – Overview and Applications in Finance

This relationship can be expressed using the inequality above.

In a positively skewed distribution, also known as a right-skewed distribution, the mean is usually to the right of the median. The mode is often less than the median, which is less than the mean.

In contrast, in a negatively skewed distribution, the mean is less than the median, which is less than the mode: $\text{mean} < \text{median} < \text{mode}$.

In a symmetrical distribution, the mean, median, and mode are all equal: $\text{mean} = \text{median} = \text{mode}$.

Question no 18:

Types of Distribution?

In statistics, common types of distributions include:

1. Binomial distribution
2. Normal distribution (also called Gaussian distribution)
3. Poisson distribution
4. Exponential distribution
5. Uniform distribution

6. Bernoulli distribution

Depend on marks you can further give details of these types.

Question no 19:

Shape of Distribution?

Answer:

The shape of a distribution of scores is usually described as either symmetric, meaning that it is similar on both sides of the center, or skewed, meaning that the values are more spread out on one side of the center than on the other. Three possible shapes: symmetric, skewed left, or skewed right.

Further it is categories into following:

1. Bimodal
2. Right-skewed
3. Bell-shaped
4. Multimodal
5. Uniform

Question no 20:

Frequency distribution?

Answer:

A frequency distribution is a way to organize data by showing the number of times a value appears in a data set.

Purpose

Frequency distributions help make data easier to understand by organizing it into a visual display.

Types

Frequency distributions can be represented as tables or graphs, and can show either the actual number of observations or the percentage of observations.

Common graphs

Histograms, bar charts, line charts, and pie charts are all graphs that can be used to represent frequency distributions.

Uses

Frequency distributions are used to represent both qualitative and quantitative data. They can be used to show election results, income, sales, and student loan amounts.

Question no 21:

If Mean is 32, median is 30 then find Mode?

Answer:

Given:

Median = 32

Mean = 30

Find:

Mode = ?

Solution:

We know that, relation between mean, median and mode can be defined as,

$$\text{Mode} = (3 \times \text{Median}) - (2 \times \text{Mean})$$

$$\Rightarrow \text{Mode} = (3 \times 32) - (2 \times 30)$$

$$= 96 - 60$$

$$= 36$$

Therefore, the value of Mode is 36.

Question no 22:

Levels or Assumption of Measurement?

Answer:

There are 4 levels of measurement, which can be ranked from low to high:

1. Nominal: the data can only be categorized.
2. Ordinal: the data can be categorized and ranked.
3. Interval: the data can be categorized and ranked, and evenly spaced.
4. Ratio: the data can be categorized, ranked, evenly spaced and has a natural zero.

Question no 23:

For a frequency distribution of variable x, mean 32, median 30, then distribution is?

Answer:

The distribution will be positive skewed distribution.

The reason is that mean is greater than median as

$$\text{Mean} > \text{Median} > \text{Mode}$$

Question no 24:

How to find Sampling probability with example?

Answer:

The formula for sampling probability is $(P = n/N)$, where (P) is the probability, (n) is the sample size, and (N) is the population. Example:

Probability formula says p of a equals number of favorable outcomes over total number of outcomes.

i.e., $P(A) = n(A)/n(S)$

Example 1: Find the probability of getting a number less than 5 when a dice is rolled by using the probability formula.

Solution

to find:

Probability of getting a number less than 5

Given: Sample space, $S = \{1,2,3,4,5,6\}$

Therefore, $n(S) = 6$

Let A be the event of getting a number less than 5. Then $A = \{1,2,3,4\}$

So, $n(A) = 4$

Using the probability equation,

$$P(A) = (n(A))/(n(s))$$

$$P(A) = 4/6$$

$$M = 2/3$$

Answer: The probability of getting a number less than 5 is $2/3$.

Question no 25:

The first and third quartile of a frequency distribution are 30 and 75. Its coefficient of skewness is 0.6. What is the median of the frequency distribution?

Answer:

According to bowley's coefficient of skewness,

$$\text{SKEWNESS} = (Q3+Q1-2\text{median})/ Q3-Q1$$

Here, given are $Q3=75$ and $Q1=30$

$$\text{So, } 0.6 = 105-2(\text{median})/45$$

$$27 = 105-2\text{median}$$

$$78 = 2\text{median}$$

Therefore, Median = 39 for the given frequency distribution.

Question no 26:

Find the Probability of 6 sided dice, and then extended it to 15 rolls, now how many contain even or odd?

Answer:

When a die is thrown, getting an even number. $\therefore$ The probability of getting an even number when a die is thrown is $\frac{1}{2}$.

There are 3 chances to roll odd numbers and 3 out of 6 of rolling even... or 50% .

Solution:

Probability of Rolling an Even Number

The probability P of rolling an even number on a single roll of the die is given by:

$P(\text{Even}) = \text{Number of even outcomes} / \text{Total outcomes}$

$= 3/6$

$= 1/2$

Rolling the Die 15 Times

When rolling the die 15 times, we can model the number of times an even number appears using a binomial distribution, where:

$N=15$ (the number of trials),

$P=1/2$

(the probability of success on each trial, which is rolling an even number).

Solution:

To find the probability of rolling an odd number on a six-sided die and then extending that to 15 rolls, we follow similar reasoning as before.

The odd numbers on a standard die are 1, 3, and 5. Thus, there are three odd outcomes out of a total of six possible outcomes.

The probability P of rolling an odd number on a single roll is $\frac{1}{2}$.

Question no 27:

Consider a 6 cards from a pack of cards without replacement, what is Probability? That three of the cards will be black?

Answer:

The probability of selecting 3 blacks from 6 cards drawn is 0.332, Can be prove by following calculation:

To find

The probability of having 3 black cards if 6 are drawn.

Explanation

Total cards available=52

Number of red cards=26

Number of black cards=26

Total ways of selecting 6 cards out of 52 cards= $52 C 6$

Total ways of selecting 3 cards out of 26 black cards = $26 C 3$.

So the probability is $(26C3 * 26C3) / 52C6$

$= \{26! / 3!(26-3)! * 26! / 3!(26-3)!\} / 52! / 6!(52-6)!$

$= \{(26 * 25 * 24) / 3 * 2 * 1\} * \{26 * 25 * 24 / 3 * 2 * 1\} / (52 * 51 * 50 * 49 * 48 * 47 / 6 * 5 * 4 * 3 * 2 * 1)$

$= 15600 / 6 * 15600 / 6 / 14658134400 / 720$

$$=2600*2600/20358520$$

$$=6760000/20358520$$

$$=0.332$$

Hence the probability of selecting 3 blacks from 6 cards drawn is 0.332

Question no 28:

Which one are Qualitative or quantitative variables:

- a). Earth weight is. _____
- b). Mass of Chocolate is 250g, then it is called _____.
- c). Number of balloons in store is _____

Answer:

All of above option statements are Quantitative variable.

Question no 29:

Committee of 5 people, consists of randomly group of 10 women, and 6 men!

Find probability? If they are arranged in:

3 women, 2 men

4women, 1 men

5 women

At least 3 women

Answer:

Committee of 5 people is to be formed randomly from a group of 10 women and 6 men, what is the probability that the committee has at least 3 women?

Answer:

Calculate the total number of ways to form a committee of 5 people from 16 individuals (10 women + 6 men).

The total number of combinations of 16 people taken 5 at a time is given by the binomial coefficient:

$$\text{Total Committees} = \binom{16}{5}$$

$$\binom{16}{5} = \frac{16!}{5!(16-5)!} = \frac{16!}{5! \cdot 11!} = \frac{16 \times 15 \times 14 \times 13 \times 12}{5 \times 4 \times 3 \times 2 \times 1} = 4368$$

1. Calculate the number of ways to form a committee with at least 3 women.

We can break this down into three cases: committees with 3 women, 4 women, and 5 women.

• **Case 1: 3 women and 2 men**

• $\text{Ways} = \binom{10}{3} \times \binom{6}{2}$

• $\binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$

• $\binom{6}{2} = \frac{6!}{2!(6-2)!} = \frac{6 \times 5}{2 \times 1} = 15$

• $\text{Total for Case 1} = 120 \times 15 = 1800$

• **Case 2: 4 women and 1 man**

• $\text{Ways} = \binom{10}{4} \times \binom{6}{1}$

• $\binom{10}{4} = \frac{10!}{4!(10-4)!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$

• $\binom{6}{1} = 6$

• $\text{Total for Case 2} = 210 \times 6 = 1260$

• **Case 3: 5 women and 0 men**

• $\text{Ways} = \binom{10}{5} \times \binom{6}{0}$

• $\binom{10}{5} = \frac{10!}{5!(10-5)!} = \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} = 252$

• $\binom{6}{0} = 1$

• $\text{Total for Case 3} = 252 \times 1 = 252$

1. Total number of favorable outcomes (at least 3 women)

$$\text{Total Favorable} = 1800 + 1260 + 252 = 3312$$

2. Calculate the probability

The probability P that the committee has at least 3 women is given by:

$$P(\text{at least 3 women}) = \frac{\text{Total Favorable}}{\text{Total Committees}} = \frac{3312}{4368}$$

To simplify:

P (At least 3 women)

≈ 0.7586

Thus, the probability that the committee has at least 3 women is approximately 0.7586, or 75.86%.

Question no 30:

Many people visit the music concert every day, If a researcher wanna get sample for those people who learning the music , then he next day get the interview of first 50 people entering in the concert!

What Type of sampling it would be?

If it is not a random sampling.

Answer:

The type of sampling described in this scenario is **Convenience sampling**.

Convenience sampling involves selecting individuals who are easiest to reach or interview, rather than randomly selecting them from the entire population.

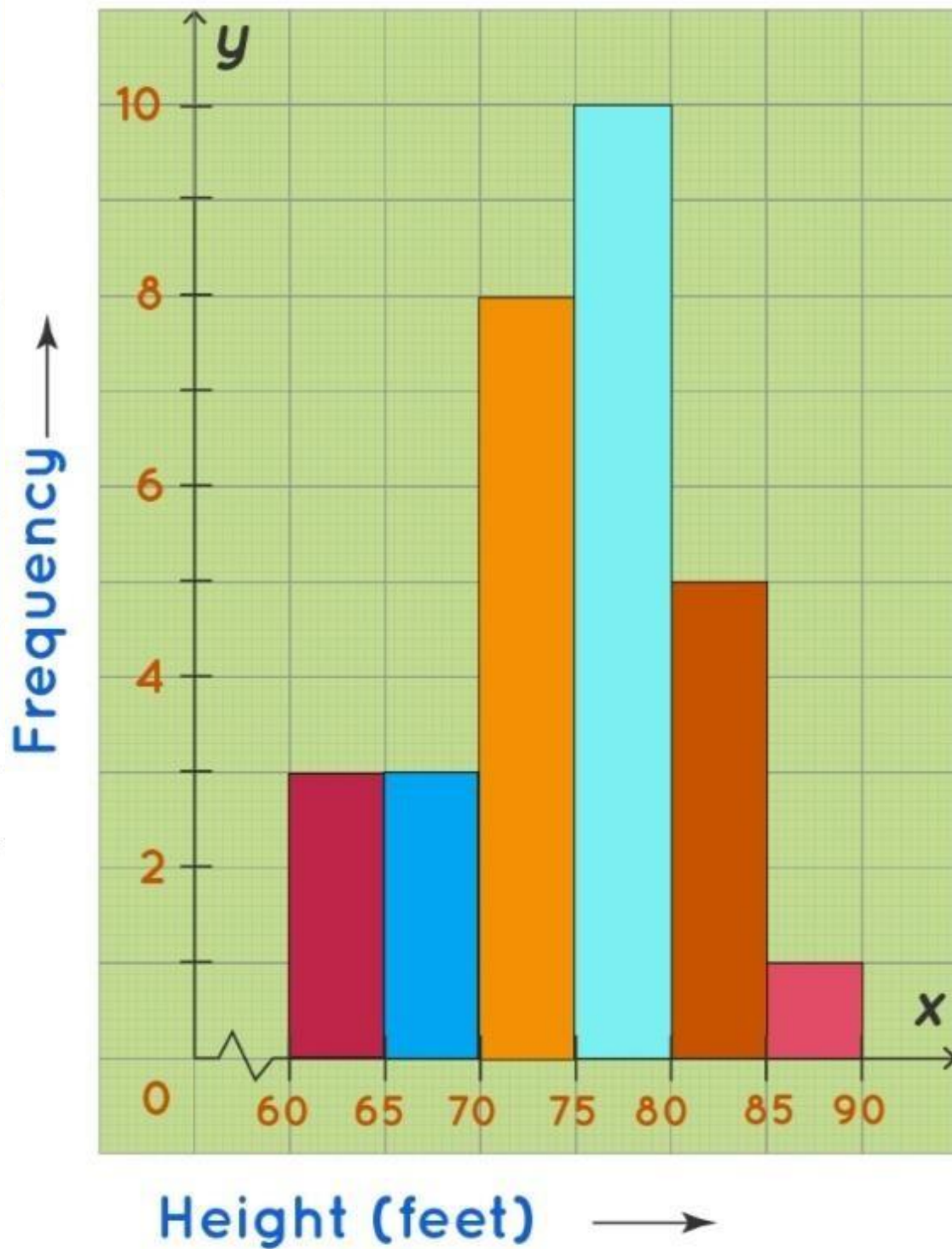
Question no 31:

Draw Histogram, if values are given?

Height Range (ft)	Number of Trees (Frequency)
60 - 75	3
66 - 70	3
71 - 75	8
76 - 80	10
81 - 85	5
86 - 90	1

This data can be now shown using a histogram. We need to make sure that while plotting a histogram, there shouldn't be any gaps between the bars.

Height of Black Cherry Trees



Note:

Method to get histogram on MS word is followed:

Tap Insert > Charts > Histogram. If necessary, you can customize the elements of the chart.

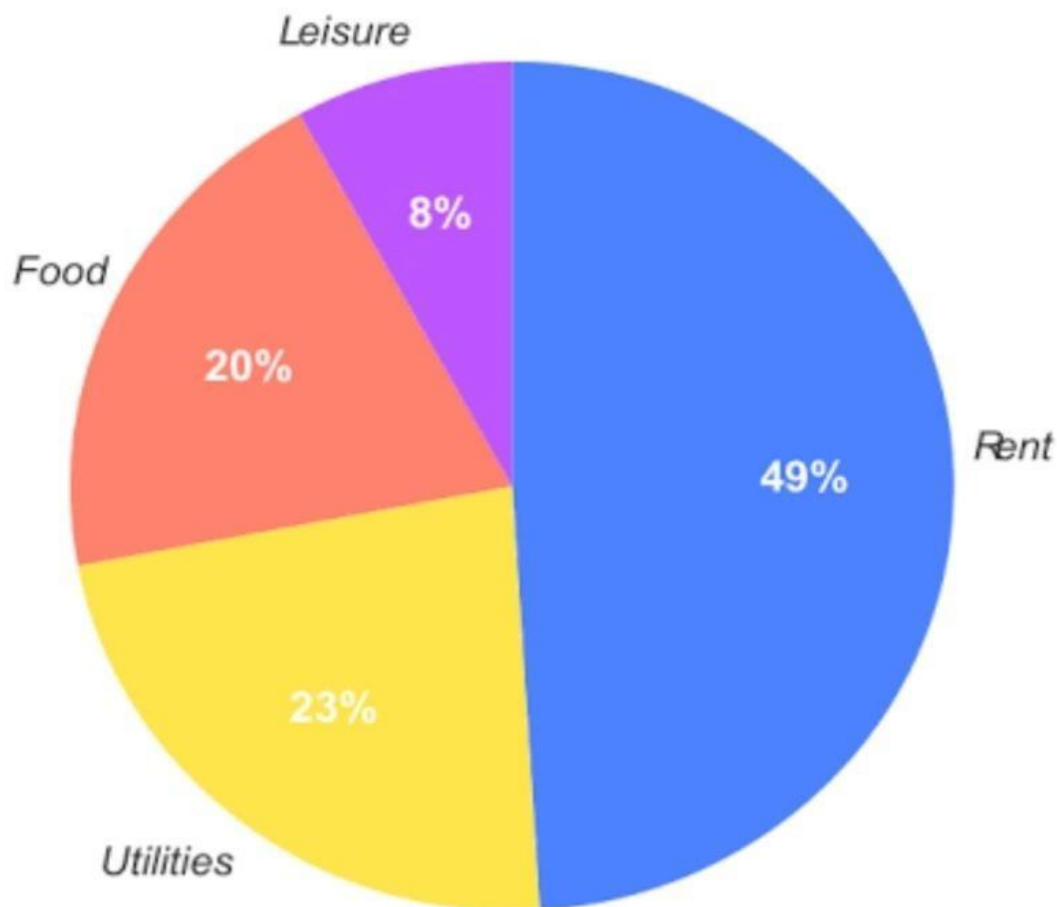
Question no 32:

Draw Pie graph if values are given?

Answer:

Same like Histogram and box and whisker plot, it's values are given and pie graph will be made by Ms word!

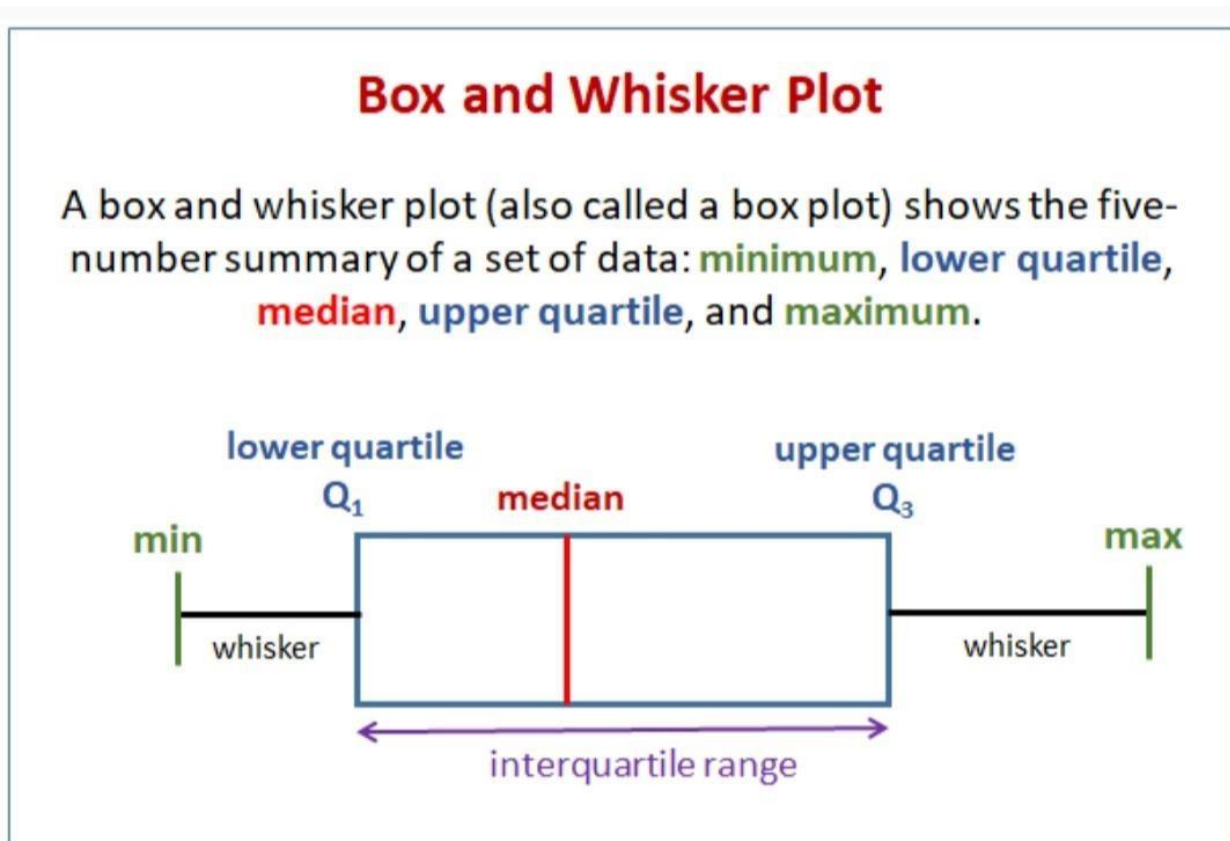
Here this how pie graph looks like!



Tap Insert > Charts > pie graph. If necessary, you can customize the elements of the chart.

Question no 33:

Whisker and Box plot diagram?



Remember:

Q₁, Q₂, Q₃, and median values will be given and you have to just make the diagram.

Except to that of diagram, sometime you have to find out the value of Q₃, if Q₁ and Q₂ values are given.

Thus, you have to prepare this whole topic properly.

Question no 34:

Box and Whisker Plot Method?

Answer:

This method is used for graphical presentation of data.

Step:

1. Arrange the data in ascending order.
2. Find mean (50% of data), lower quartile (25%), upper quartile (75%) of the data.
3. Draw a number line that include smallest and largest value of the data.
4. Draw three vertical lines at the lower quartile, mean and upper quartile above the number Line.
5. Join the lines for lower quartile and upper quartile to form a box
6. Draw the line from box at left side for smallest value and a line from box to right for largest Value.
7. The difference between upper and lower quartile is called interquartile range (IQR).

Video explanation:

<https://youtu.be/7jP3TowC9U4?si=4BJoYzmpV5zHt7KH>

Question no 35:

1. If Mean is 35, Median is 40 and Mode is 41, then what kind of its distribution will be? OR another example is:
2. For a distribution, mean is 40, median is 40.5 and mode is 41. The distribution is:

Answer:

So! In both above given examples, median is greater than mean, and mode is greater than both (mean and median).

Solution:

Given:

Mean = 40

Mode = 41

Median = 40.5

In negative skewed = Mean is minimum while mode is maximum and median is between mean and mode.

Here Mean < Median < Mode.

∴ The distribution is negatively skewed.

Note:

If mean greater than median, then the distribution will be positive skewed distribution!

Question no 36:

Define Variance with formula in Hypothetical form?

Answer:

It is calculated by taking the differences between each number in the data set and the mean, squaring the differences to make them positive, and then dividing the sum of the squares by the number of values in the data set. Software like Excel can make this calculation easier.
(This definition is in hypothetical form)

Variance Example:

The mean is given as $(3 + 5 + 8 + 1) / 4 = 4.25$. Then by using the definition of variance we get $[(3 - 4.25)^2 + (5 - 4.25)^2 + (8 - 4.25)^2 + (1 - 4.25)^2] / 4 = 6.68$. Thus, variance = 6.68.

Question no 37:

Table is given below:

Count	Frequency
1	5
3	5
5	6
7	1
11	8

Find the Cumulative frequency?

The cumulative frequency is calculated by adding each frequency from a frequency distribution table to the sum of its predecessors.

Question no 38:

A researcher team is conducting a study on the exercise habits of adults in a small town . The team has 2000 adults and the team want to survey 200 adults together the data. They decide randomly select 200 hundred names from the list of all 2000 adults in a town using computer generated random sampling?

Answer:

Yes, if the researcher team uses computer-generated random sampling to select 200 names from a list of 2,000 adults, they are using **simple random sampling**. Simple random sampling is a probability method where each member of a population has an equal chance of being selected.

Question no 39:

In the group of 40 people, 10 are healthy and other people of remaining as 30 have either High blood pressure, High level of cholesterol or Both, if 15 have High blood pressure and 15 have High level of cholesterol, then find the -----

Count	Frequency	Cumulative Frequency	Cumulative Relative Frequency
1	5	5	$\frac{5}{25} = 0.2$ 20%
3	5	10	$\frac{10}{25} = 0.4$ 40%
5	6	16	$\frac{16}{25} = 0.64$ 64%
7	1	17	$\frac{17}{25} = 0.68$ 68%
11	8	25	$\frac{25}{25} = 1.0$ 100%

8. Has high Blood pressure (Event A)
9. Has high Level of cholesterol (Event B)

OR

In a group of 40 people, 10 are healthy and every person of the remaining 30 has high blood pressure, a high level of cholesterol or both. If 15 have high blood pressure and 25 have high level of cholesterol, how many people have high blood pressure and a high level of cholesterol?

Answer:

Let's define A = "People with high blood pressure" and B = "People with high cholesterol." Since 10 of the 40 are healthy and have neither high blood pressure nor high cholesterol, we can effectively disregard those 10 and focus on the remaining "unhealthy 30."

The problem states that $A \cup B = 30$ (that reads "A union B equals 30"... I'm not sure how to insert the actual math symbol for union). Also, $A = 15$ and $B = 25$.

Some people have both conditions. These people are members of the set $A \cap B$ ("A intersection B"), or in other words, people with both high blood pressure and high cholesterol.

Note that $A \cup B = A + B - A \cap B$. Since we are looking for $A \cap B$, we can rearrange both equations to get the following: $A \cap B = A + B - A \cup B$.

Thus:

Plug and chug: $A \cap B = 15 + 25 - 30 = 40 - 30 = 10$, the final answer.

Question no 40:

If Minimum and Maximum values are given, then find the Median?

Answer:

Let's say you have a set of numbers: 5, 10, 13, 1, 5, 8

The maximum would be the greatest number in the set: 13

The minimum would be the smallest number in the set: 1

The mean is the average: 7 ($5+10+13+1+5+8=42$, $42/6=7$)

Question no 41:

Detail on Mean, Mode Median and Range?

The mean, or average, is calculated by summing up all the values in a dataset and dividing the sum by the number of values.

Example:

Consider the dataset {12, 15, 18, 22, 25}.

Mean = $(12 + 15 + 18 + 22 + 25) / 5 = 92 / 5 = 18.4$

Mode:

The mode is the value that appears most frequently in a dataset.

Example:

For the dataset {5, 8, 12, 8, 15, 8, 22}, the mode is 8 because it appears more frequently than any other value.

Median:

The median is the middle value when a dataset is ordered. If the dataset has an even number of values, the median is the average of the two middle values.

Example:

For the dataset {10, 15, 18, 22, 25}, the median is 18. For the dataset {10, 15, 18, 22}, the median is $(15 + 18) / 2 = 16.5$.

Range:

The range is the difference between the maximum and minimum values in a dataset.

Example:

For the dataset {8, 12, 5, 15, 22}, the range is $22 \text{ (maximum)} - 8 \text{ (minimum)} = 14$

Question no 42:**Branches/ Types of Biostatistics:****1. Inferential Statistics:**

Inferential statistics, on the other hand, go beyond the immediate data and use probability theory to make inferences or predictions about a population based on a sample of data. This includes techniques like hypothesis testing, confidence intervals, and regression analysis. Inferential statistics help draw conclusions and make generalizations about populations by analyzing a subset of the data.

2. Descriptive Statistics:

Descriptive statistics involve methods for summarizing and presenting data in a meaningful way. This includes measures of central tendency like the mean, median, and mode, as well as measures of dispersion like range and standard deviation. Descriptive statistics aim to provide a clear and concise overview of the main features of a dataset.

- Descriptive statistics is performed using Tabular and Graphical Methods.

- i) **Tubular Method:** It is used to summarize the data in table form. The most frequent method is used in tabular form is frequency table.
- ii) **Graphical Methods:** It is visual way of parenting data using charts and graphs. The most frequent methods are used for this purpose are Histogram, Box, Whisker Plot, Scatter plot etc.

Question no 43:**Frequency Distribution?**

Frequency refers to the number of times a particular value occurs in a dataset and Frequency distribution is a method used to organize and summarize the data in tables and give more insight of the data.

There are three types of frequency distribution.

1. **Categorical Frequency Distribution:** It is used for qualitative data where categories have no any

Order such as colors, gender etc.

2. **Ungrouped Frequency Distribution:** It is used for the quantitative data mainly dealing with small dataset and each value is treated separately.
3. **Grouped Frequency Distribution:** Also used for quantitative data but deals with larger data set where data grouped into many interval or classes.

Relative Frequency = frequency of the class / Total

Cumulative Relative Frequency = sum of previous relative Frequencies + current class frequency.

In categorical and ungrouped frequency distribution we make a table. Write the data (avoid repetition) and use Tally Marks for counting and write frequency in 3rd column. We can also find relative frequency and Cumulative Relative Frequency in 4th and 5th columns using above formulas.

Steps for Grouped Frequency Distribution:

1. Write the data in order array.
2. Find the smallest value (X_o) and largest value (X_m)
3. Find the range which is difference between largest and smallest value.
4. Find width (class interval) denoted by h : divide range by number of chosen classes (Often 5 to 15 classes to be assumed).
5. Make number of classes from smallest value to largest value based on class interval.
6. Now draw tally mark and find the frequency of the dataset.

Frequency Histogram:

After the above process if we make bars for visual display of the data then it is called frequency histogram. The height of the bar represent the frequency. Best video for frequency distribution explanation:

<https://youtu.be/riYpfCIuOUo?si=EsjjuDC4qWKHF5ek>

Question no 44 :

Permutations and Combinations

The arrangements of objects in a specific order.

Case 1:

When all objects are distinct:

Formula: $nPr = n!/(n-r)!$

Example:

A club consists of four members. How many ways are there of selecting three officers: president, Secretary and treasurer? It is evident that the order, in which 3 officers are to be chosen, is of Significance.

$N = 4$

$R = 3$ Put

in formula:

$4P3 = 4! / (4 -$

$3)! 4 \times 3 \times 2 \times 1 =$

24 Case 2:

When some objects are repeated:

Formula:

$P = n! / n_1! \times n_2! \times n_3! \dots$

Example:

Find number of permutations in the word "BOOKKEEPER,"

- Total number of letter (n) = 10

- O = 2

- K = 2

- E = 3

$P = 10! / 2! \times 2! \times 3!$

$P = 151200$

Combination :

The selections of elements without considering the order.

Formula: nCr

$= n! / r! (n-r)!$

Example:

A group of 5 people, and you want to form a committee of 2 people. Find number of Combination.

$N = 5$

$R = 2$

$C = 5! / 2! (5-2)$

$C = 10$

combinations

Video

Explanation:

Short video : https://youtu.be/GZATQeI8dJo?si=H_IXfmUA7GrShg6y

Long video complete concept: <https://youtu.be/Tr-TVt5JAWY?si=3Ozrz9wWSERRsnxZ>

Question no 45:

Mean Deviation

Answer:

Example : Set of data: [2, 4, 6, 8, 10]. Find mean deviation.

1. Calculate the Mean:

$$2 + 4 + 6 + 8 + 10 / 5 = 6$$

2. Find the Deviation for Each Data Point:

- Deviation for 2: $(2 - 6 = -4)$
- Deviation for 4: $(4 - 6 = -2)$
- Deviation for 6: $(6 - 6 = 0)$
- Deviation for 8: $(8 - 6 = 2)$
- Deviation for 10: $(10 - 6 = 4)$

3. Sum of Deviations: [

$$(-4) + (-2) + 0 + 2 + 4 = 0]$$

Variance:

1. Square the mean deviation and then sum up and divide by number of dataset.

$$16 + 4 + 0 + 4 + 16 / 5 = 8$$

Standard deviation Standard

Deviation:

Take the square root of variance value

$$\sqrt{8} = 2.83$$

Coefficient of standard deviation

Coefficient of standard deviation = Standard deviation / Mean x 100

$$2.83 / 6 \times 100 = 47.16$$

2.84

Video Explanation:

https://youtu.be/tXv1KVoQrWA?si=y_ZS0-JQSc62x5d

<https://youtu.be/179ce7ZzFA8?si=77AQC7y8qFUGmlqr>

Question no 46:

Probability

Answer:

Probability = Number of favorable outcomes/ Total number of possible outcomes

Example:

Tossing a fair six-sided die. Find the Probability of Rolling an Even Number:

Possible outcomes = 2,4,6

Total Sides = 6

$$P = 3/6 = 1/2$$

Video Explanation:

<https://youtube.com/shorts/6oCvo2BULbU?si=JEx8UtCxynvw889l>

BIO401 Experts Zoologists Group link:
<https://chat.whatsapp.com/EZweNC0owIb0MiKly9ikrn>

Regards:

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